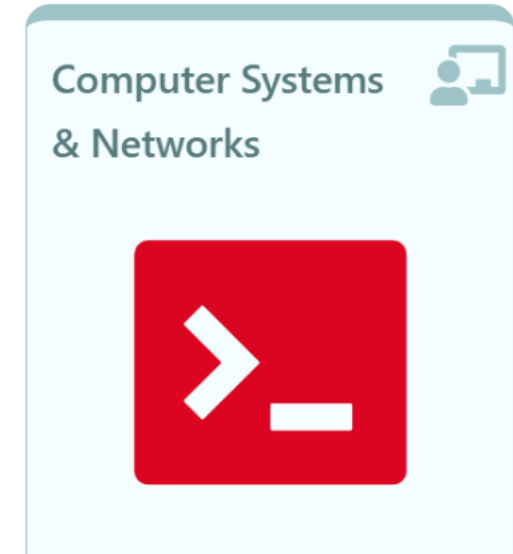
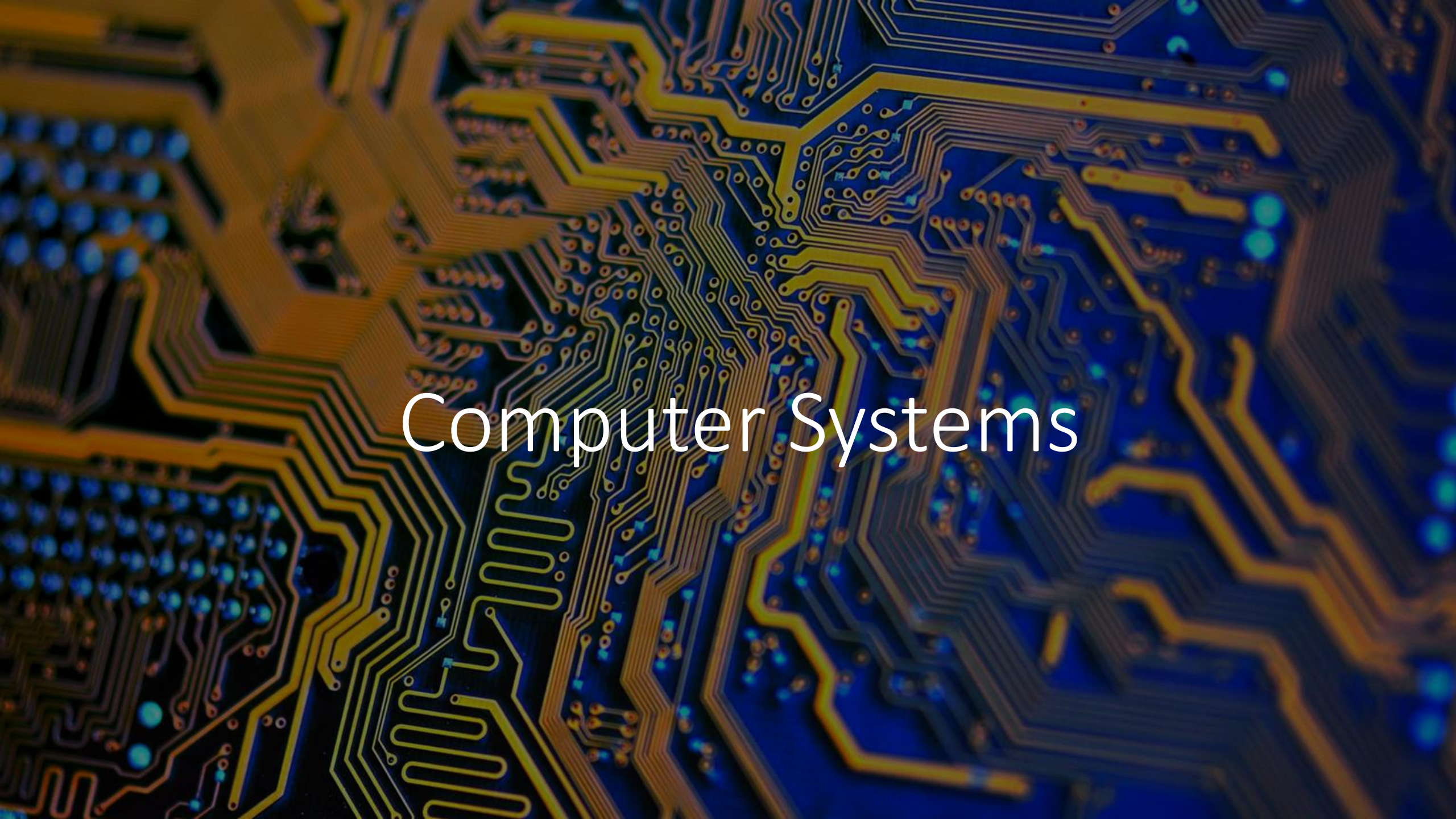


Computer Systems & Networks





Computer Systems

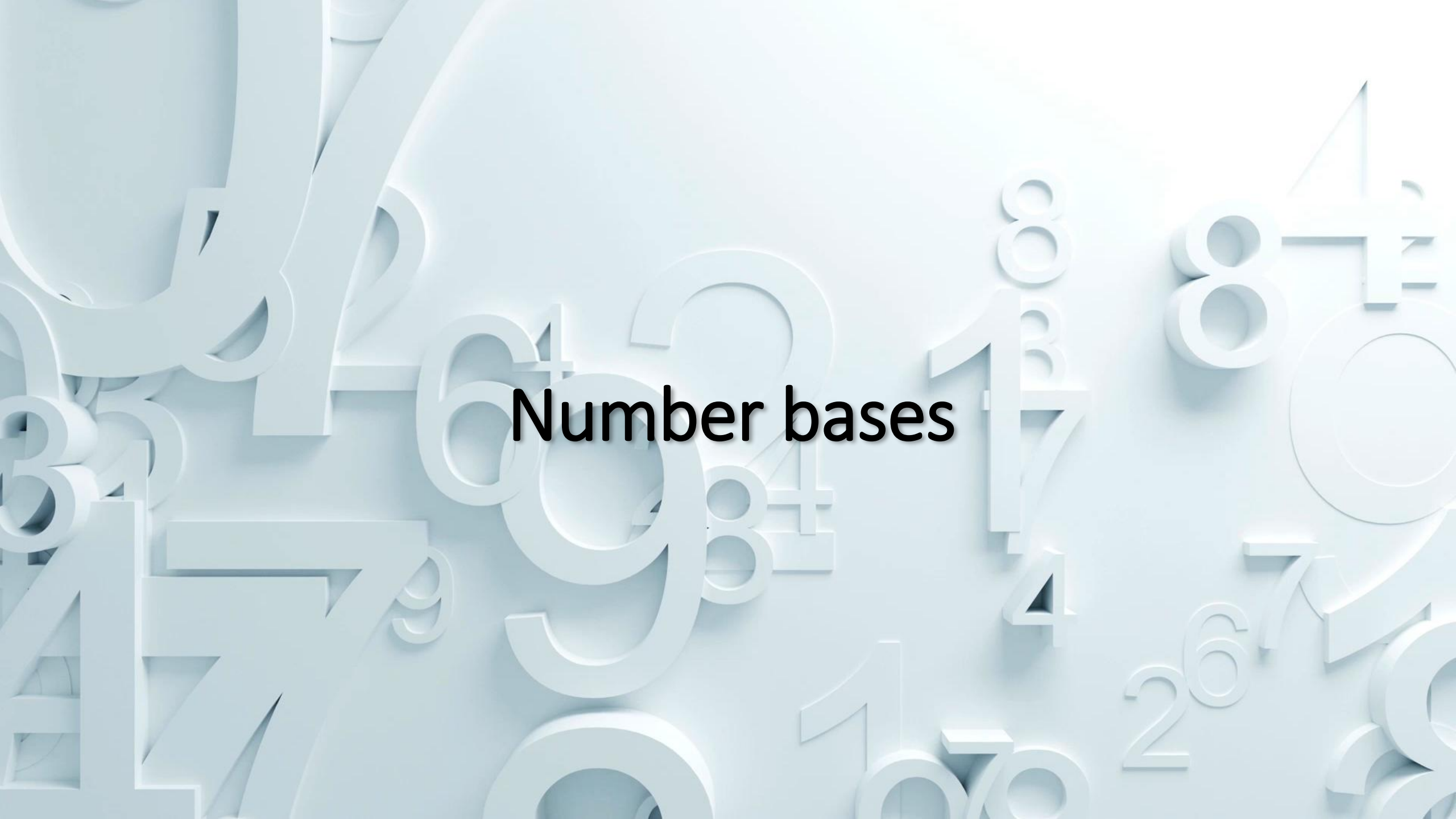
- 1) Memory was something that you lost with age
- 2) An application was for employment
- 3) A program was a TV show
- 4) A cursor used profanity
- 5) A keyboard was a piano!
- 6) A web was a spider's home
- 7) A virus was the flu!
- 8) A CD was a bank account
- 9) A hard drive was a long trip on the road
- 10) A mouse pad was where a mouse lived



```
compsys@compsys-virtualbox:~$ ./stats  
bash: ./stats: Permission denied
```

There are basically **four** major types of number systems, namely:

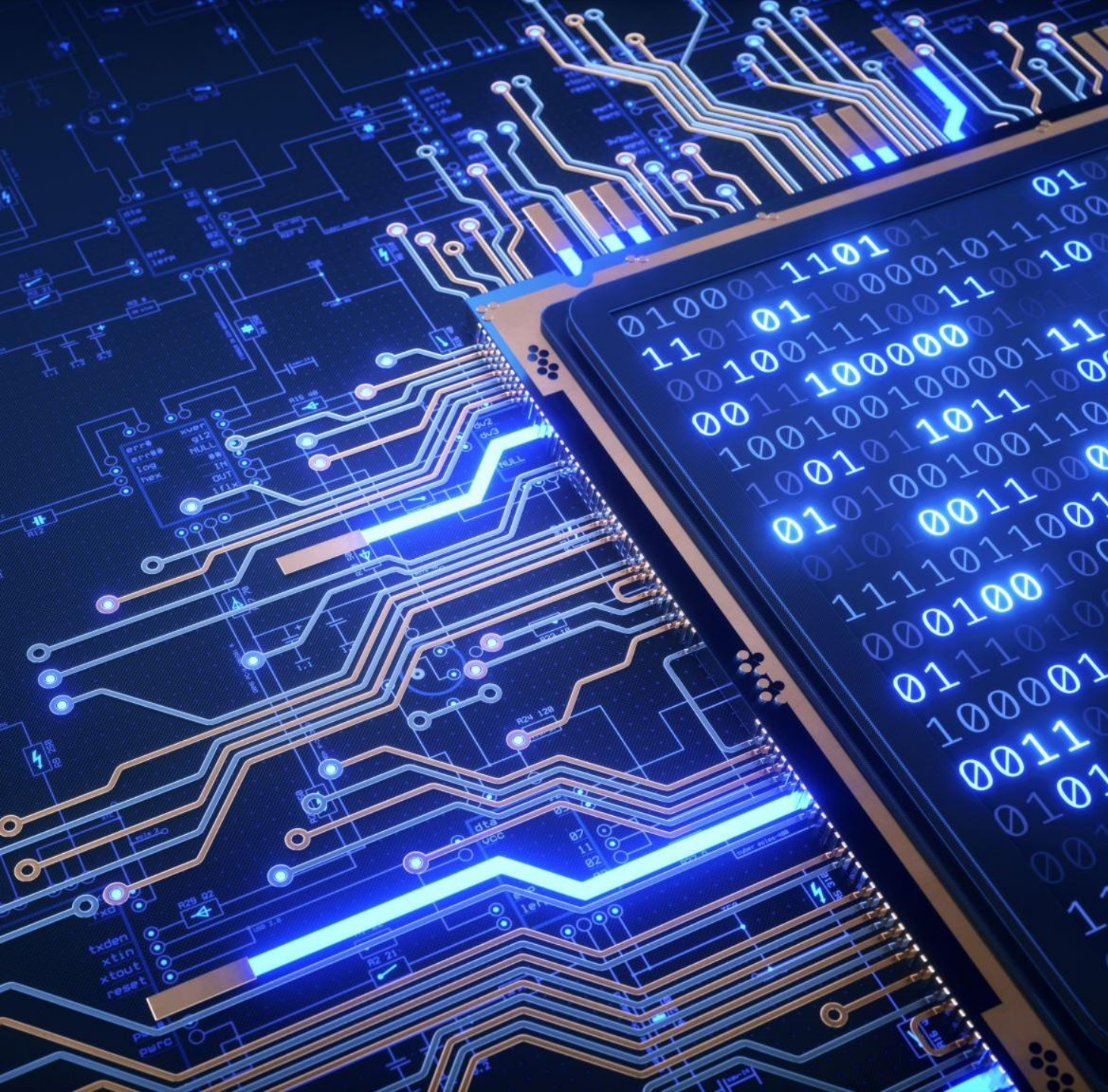
1. decimal (base10),
2. binary (base2),
3. octal (base8) and
4. hexadecimal (base16)

The background is a light blue gradient with various 3D-rendered numbers and mathematical symbols scattered across it. These include digits from 0 to 9, the infinity symbol (∞), and the percent sign (%). The numbers are of different sizes and are cast with soft shadows, giving them a three-dimensional appearance as if they are floating or resting on a surface.

Number bases

Number Bases in Computer Systems

Decimal (base 10)	Binary (base 2)	Octal (base 8)	Hexadecimal (base 16)
00	0000	00	0
01	0001	01	1
02	0010	02	2
03	0011	03	3
04	0100	04	4
05	0101	05	5



1 of 4:
Decimal

- Let's analyze the decimal number **375**:

$$375 = (3 \times 100) + (7 \times 10) + (5 \times 1)$$

$$= (3 \times 10^2) + (7 \times 10^1) + (5 \times 10^0)$$

Position weights $\longrightarrow$ **10^2 10^1 10^0**

Number digits $\longrightarrow$

3 7 5

$$5 \times 10^0 = 5 +$$

$$7 \times 10^1 = 70 +$$

$$3 \times 10^2 = 300$$

375

1	8	2	7	3	6
10^5	10^4	10^3	10^2	10^1	10^0
					$6 \times 10^0 = 6$
				$3 \times 10^1 = 30$	
			$7 \times 10^2 = 700$		
		$2 \times 10^3 = 2000$			
	$8 \times 10^4 = 80000$				
$1 \times 10^5 = 100000$					
					182736

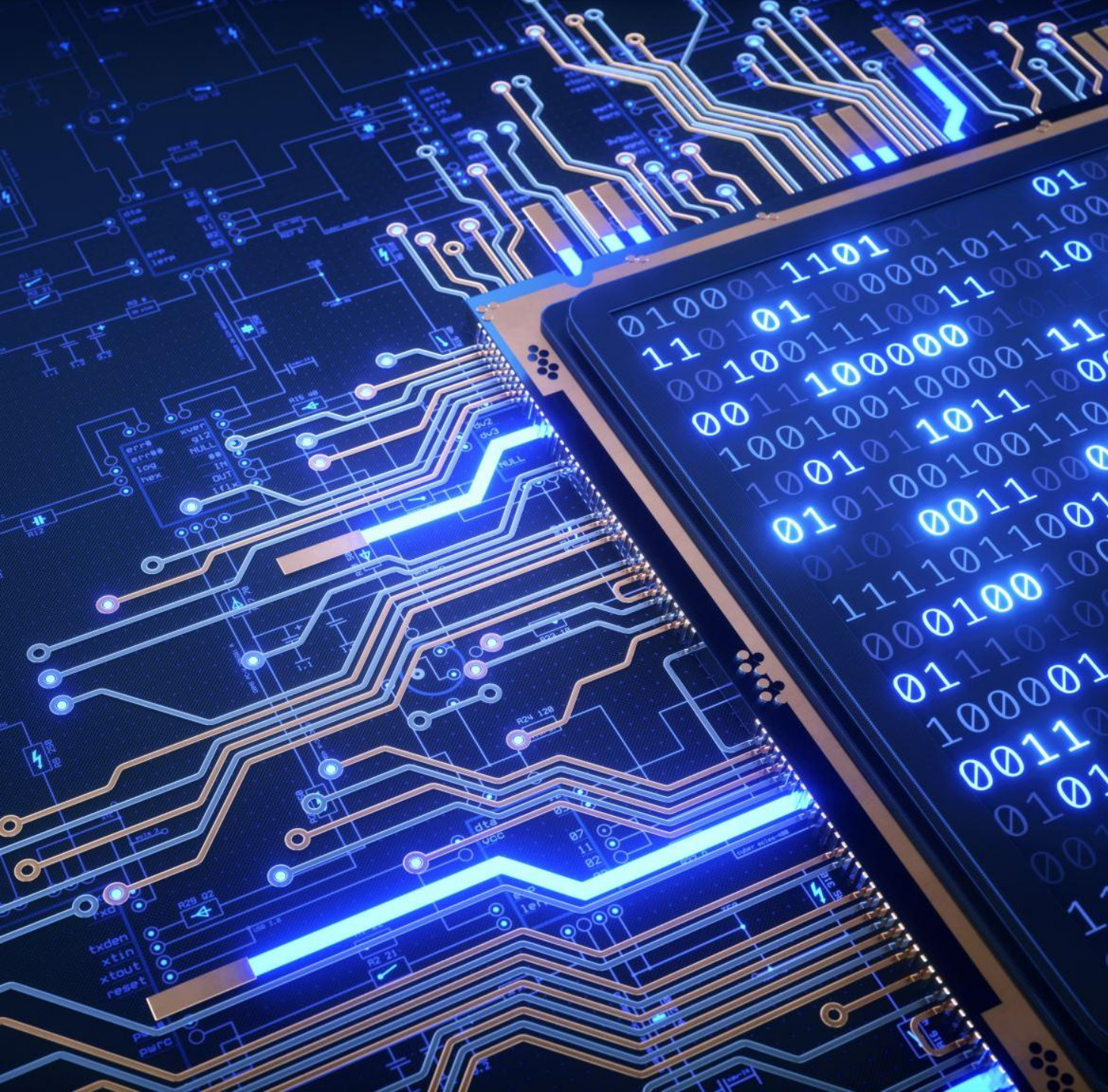
... 100000 10000 1000 100 10 1

... 10^5 10^4 10^3 10^2 10^1 10^0

....654321

Sixth digit Fifth digit Fourth digit Third digit Second digit First digit

value of digit in "Decimal numeral system"



2 of 4:
Binary

1 inputs: binary

- For 1 input there are 2^1 possible outputs
- Represents either being on/off



INPUT 1: Light	POSSIBLE OUTPUTS:
0	Light off
1	Light on

(1) NUMBER BASE: Binary

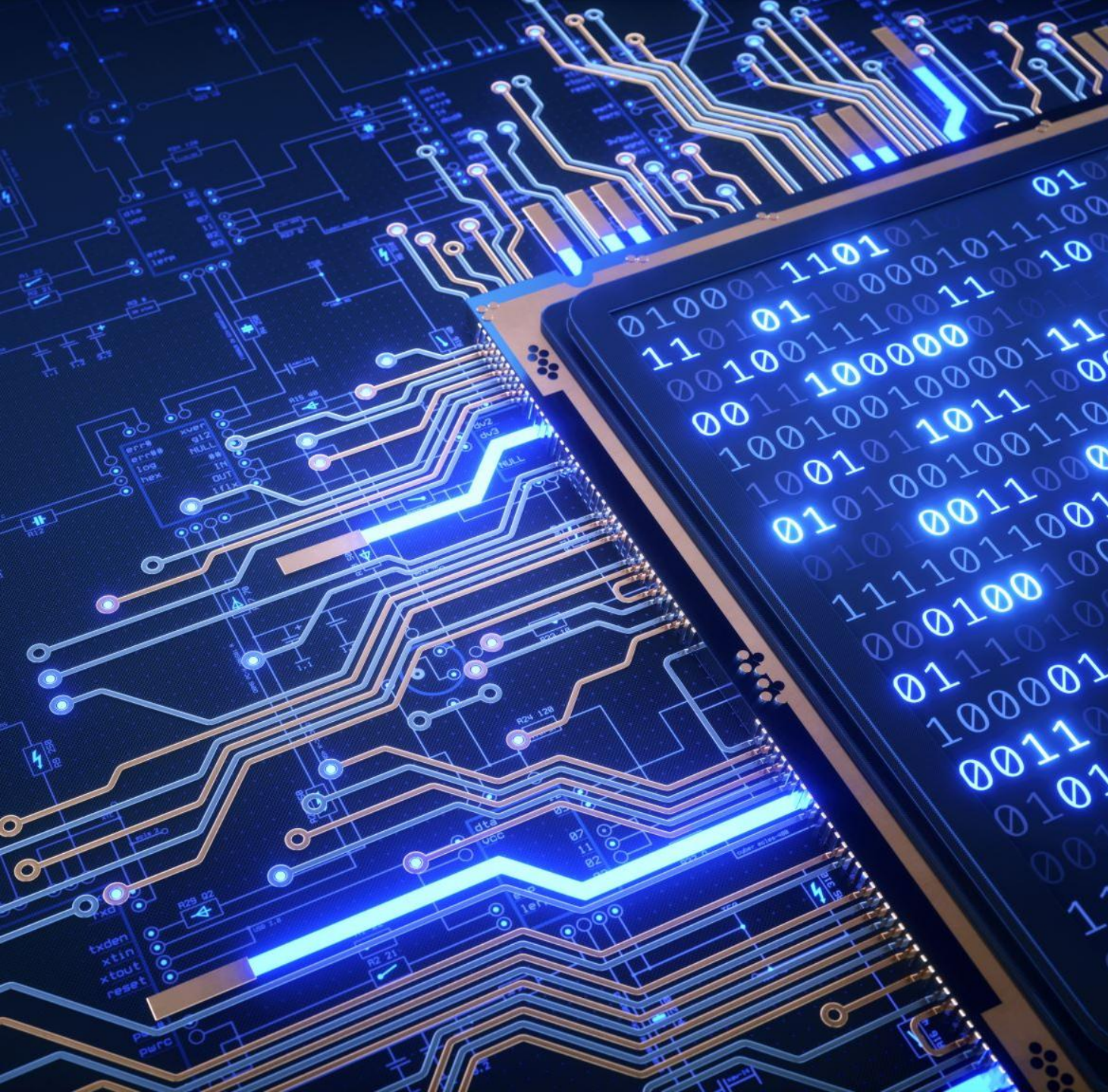
- a base-2 numeral system that uses two symbols (typically represented as) 0 and 1, to represent numeric values

	Column 8	Column 7	Column 6	Column 5	Column 4	Column 3	Column 2	Column 1
<u>Base</u> ^{exp}	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
Weight	128	64	32	16	8	4	2	1

Sample Storage amounts:

1 TB
512 GB
256 GB
128 GB
64 GB
16 GB
32 GB

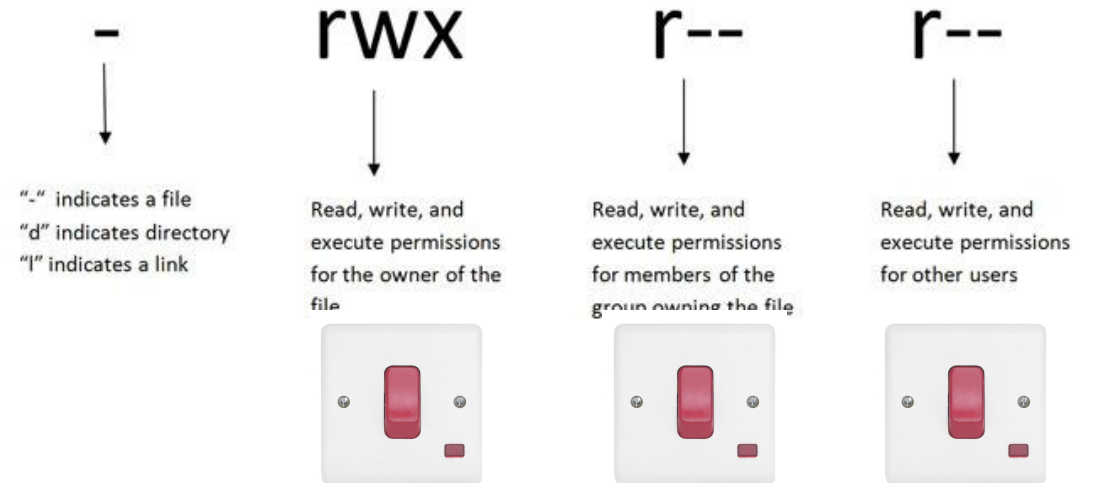
	Column 8	Column 7	Column 6	Column 5	Column 4	Column 3	Column 2	Column 1
<u>Base</u> ^{exp}	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
Weight	128	64	32	16	8	4	2	1



3 of 4:
Octal

Octal Representation

0	000	- - -	No permissions
1	001	- - x	Only Execute
2	010	- w -	Only Write
3	011	- w x	Write and Execute
4	100	r - -	Only Read
5	101	r - x	Read and Execute
6	110	r w -	Read and Write
7	111	r w x	Read, Write and Execute



(2) NUMBER BASE: Octal

- base-8 numeral system
 - uses eight symbols, typically represented as digits 0 through 7, to represent numeric values
- Each digit's position in an octal number represents a power of 8
- distinct value of between $000_2(0_8)$ and $111_2(7_8)$ Why?

Change Permissions

File(s):

/public_html/wp-content

Mode	User	Group	World
Read	☒	☒	☒
Write	☒	☒	☒
Execute	☒	☒	☒
Permission	7	7	7



Binary	Octal
0000	0
0001	1
0010	2
0011	3
0100	4
0101	5
0110	6
0111	7
1000	10
1001	11
1010	12
1011	13
1100	14

- In the octal representation, each permission is represented by a digit:

4 for Read,
2 for Write, and
1 for Execute

Why?

File Permissions

Octal	7	5	5 (base8)
≡Binary	111	101	101 (base2)
	Owner	Group	Others

Show that $7_8 \equiv 7_{10}$

Weighting	8^3	8^2	8^1	8^0
=	512	64	8	1
No. to Convert:				7_8
				$+(7 \cdot 8^0)$ $+7 \cdot 1$ $+7$
			$7_8 = 7_{10}$	

Show that $75_8 \equiv 61_{10}$

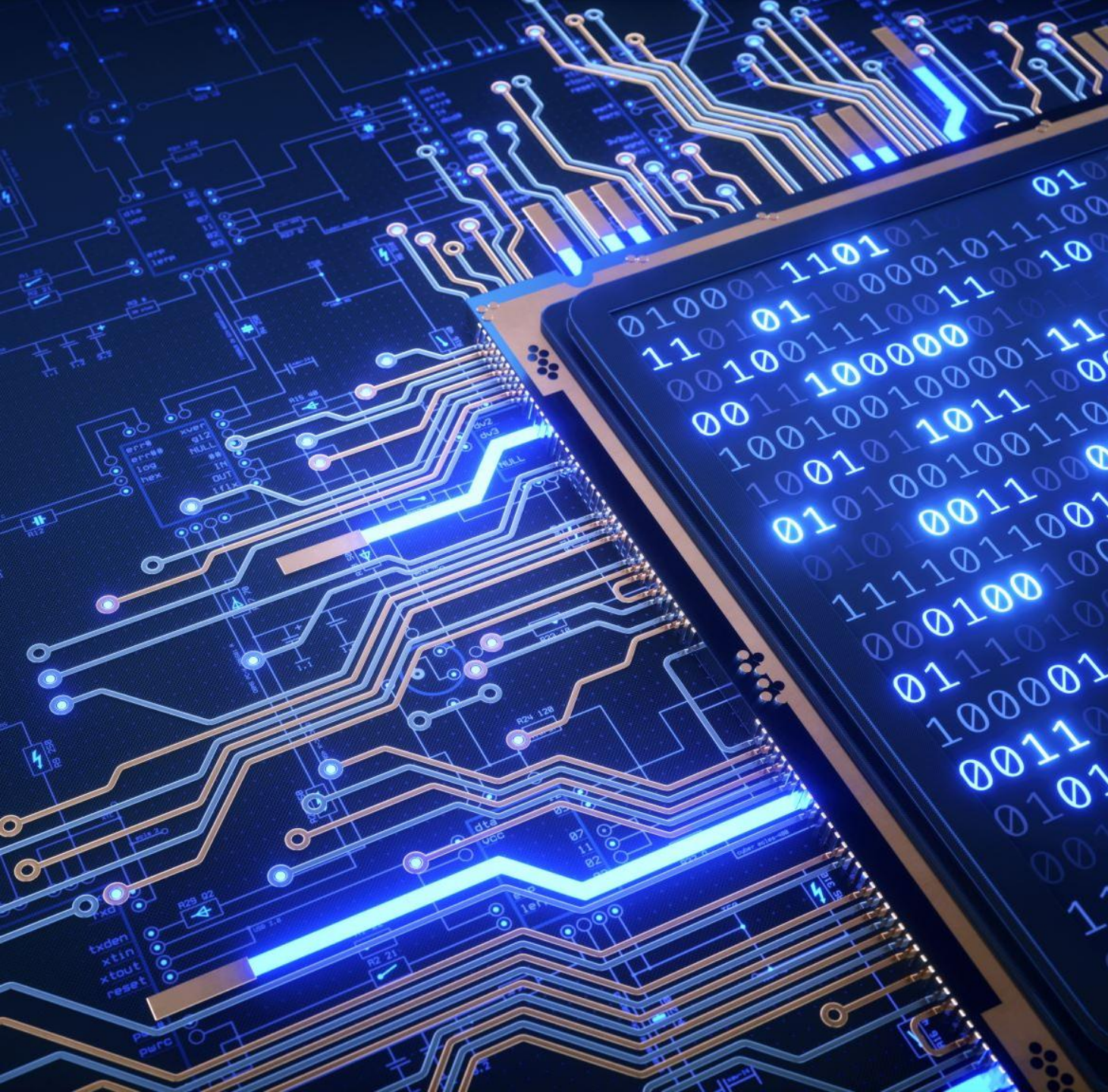
Weighting	8^3	8^2	8^1	8^0
=	512	64	8	1
No. to Convert:			7_8	5_8
			(7×8^1) $= 7 \times 8$ $= 56$	$+(5 \times 8^0)$ $+ 5 \times 1$ $+ 5$
			$75_8 = 61_{10}$	

chmod

is used to
change
permissions of
a file

CHMOD is used to change permissions of a file.

PERMISSION			COMMAND
U	G	W	
rwX	rwX	rwX	chmod 777 filename
rwX	rwX	r-X	chmod 775 filename
rwX	r-X	r-X	chmod 755 filename
rw-	rw-	r--	chmod 664 filename
rw-	r--	r--	chmod 644 filename
User	Group	World	r = Readable w = Writable x = Executable - = None



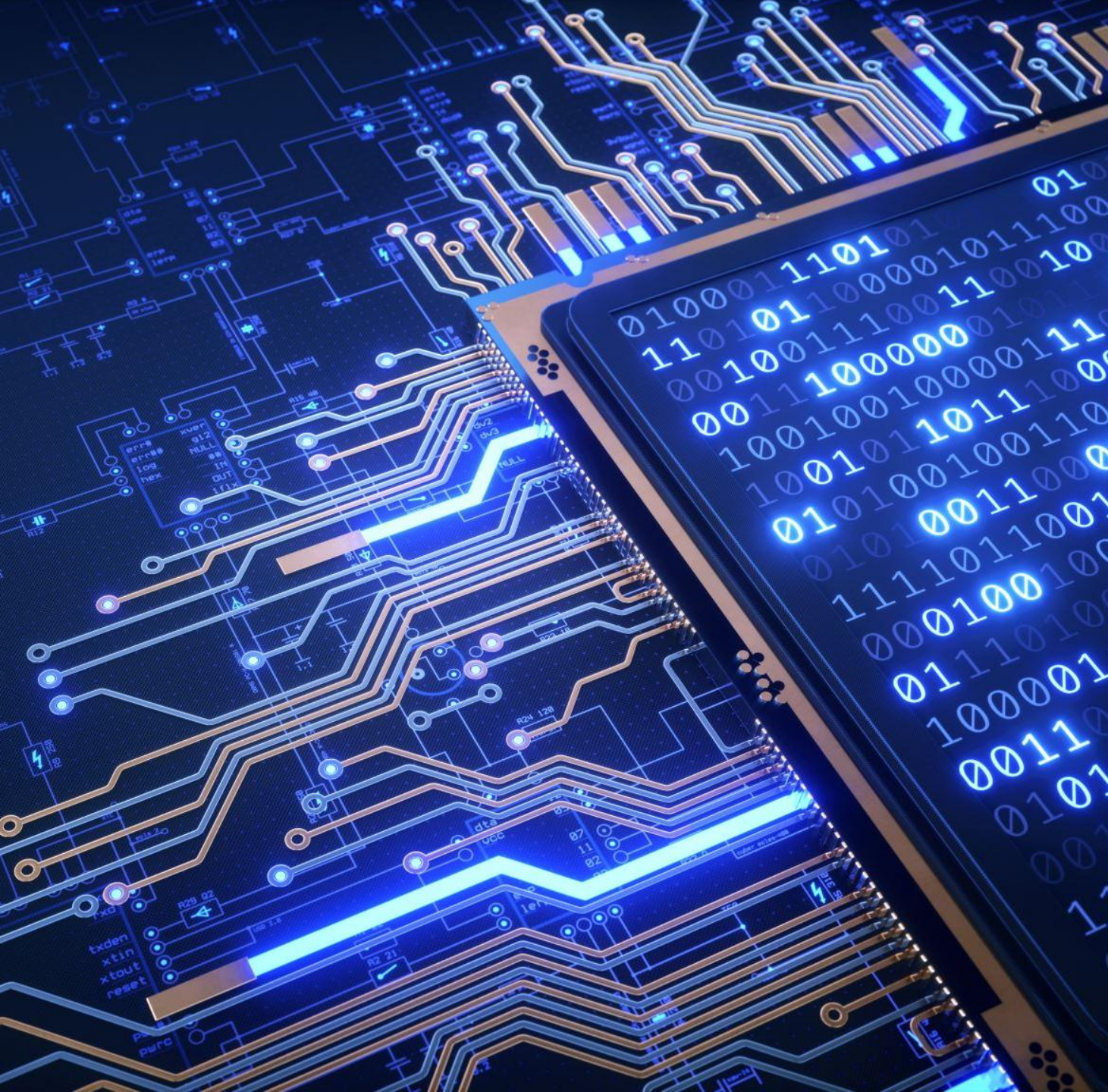
4 of 4:
Hexadecimal

(3) NUMBER BASE: Hexadecimal (later in sem)

- Longer binary numbers are what computers use at the hardware level which is not human-user-friendly.
- Hex uses 16 symbols to represent its value:

Hexadecimal Symbol	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
Decimal Value	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15

Color Name	Color Code
Red	<u>#FF0000</u>
Cyan	<u>#00FFFF</u>
Blue	<u>#0000FF</u>
DarkBlue	#0000A0



In Summary

Octal is:

... a concise way to set and view Linux file permissions numerically

- where each digit represents the access level for:

1. owner,
2. group, and
3. others

respectively.

Three categories of file user

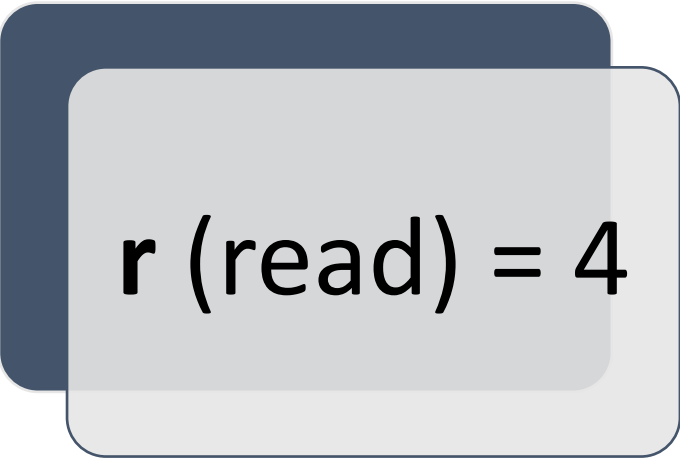
Owner = permissions
represented in the first
octal digit

2. Group = permissions
represented in the
second octal digit

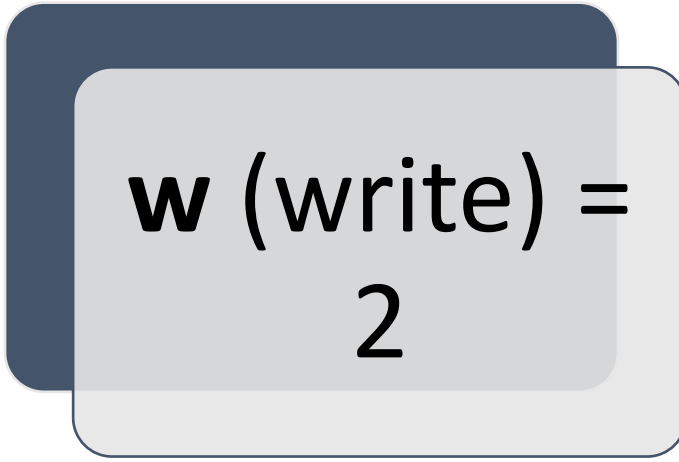
3. Others =
permissions
represented in the
third octal digit



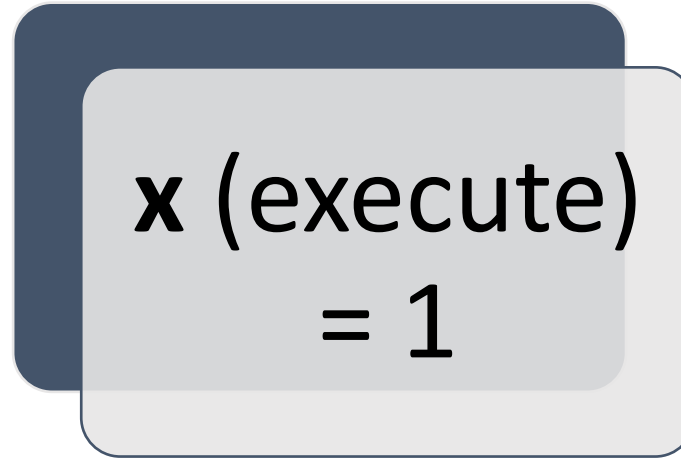
Each set has **three** types of permissions



r (read) = 4



w (write) =
2



x (execute)
= 1

These values are **added together** for each set

So a permission:

`rwXr-Xr-`

becomes:

Owner: $rwX = 7$

Group: $r-X = 5$

Others: $r-- = 4$

Octal: 754(base 8) written as 754_8

Try out today's lab

01: Computer Systems



NetAcad · Linux as an OS

02: File Permissions

	rw	r--	r--
-	Read, write, and execute permissions for the owner of the file	Read, write, and execute permissions for members of the group owning the file	Read, write, and execute permissions for other users

Numbering Systems · Permissions · Octal

Lab



Fundamentals · Commands
· Scripts · Permissions

File Hierarchy and Navigation

Creating Files, Absolute and Relative Filenames

Navigation Basics

Redirection

File Manipulation

Executable Scripts

Week 1 Summary

Cisco NetAcad